

Sanchay Narendra Gawande

sanchay823@gmail.com | (857) 395-5752 | Boston, MA, USA | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

Machine Learning and Software Engineer with 2+ years of experience building production ML and AI systems in Python, from data pipelines and evaluation to deployment and monitoring on AWS. Hands on experience with LLM workflows including RAG, tool calling, and low latency inference services using FastAPI, Docker, and PostgreSQL.

Technical Skills

- Programming Languages: **Python**, JavaScript (Node.js, React, Redux), **SQL**, Java
- Backend Frameworks and APIs: Flask, **FastAPI**, Node.js, GraphQL, RESTful APIs
- Frontend Development: React, Redux, Next.js, D3.js, HTML5, CSS3
- Databases and Data Storage: **PostgreSQL**, MongoDB, Redis, Firebase Realtime Database
- Cloud Platforms and Services: **AWS** (Lambda, API Gateway, S3, EC2, CloudFront, ECS), Serverless Architecture, **Docker**, Kubernetes, GitHub Actions
- Version Control and Collaboration: Git, GitHub, Pull Request workflows, CI/CD, API design principles, Microservices, Serverless architecture
- Machine Learning and AI Tools: **PyTorch**, **scikit learn**, XGBoost, LightGBM, TensorFlow Serving, ONNX Runtime, OpenAI API and embeddings, DistilBERT
- Agents and Orchestration: LangChain, LangGraph, tool execution, function calling

Professional Experience

Machine Learning & AI Engineer (Founding Team), Suno Analytics 01/2025 – Present | Remote, USA

- Architected the core multi model LLM orchestration layer using **FastAPI**, serving as the primary engineering owner for the ERP platform AI capabilities from design to production.
- Designed a stateful RAG system using **Pinecone** and metadata driven routing, enabling long running AI assistants to maintain context across complex financial analysis workflows.
- Engineered an asynchronous ingestion pipeline using **AWS Lambda** and **SQS** to merge multi tab Excel workbooks, reducing user facing ingestion latency by 40% compared to the legacy synchronous process.
- Built a tool calling infrastructure (MCP server) integrating Claude and GPT 4 pipelines, enabling the AI to trigger real time actions like macro generation.
- Defined retrieval quality and response regression checks for RAG workflows, improving iteration speed and reducing production debugging time during prompt and routing updates.
- Optimized **AWS** infrastructure (**ECS**, API Gateway) for cost and reliability, establishing monitoring for P99 latency across the orchestration layer.

Software Engineer Fellowship, Headstarter 07/2024 – 01/2025 | Remote, USA

- Built multiple full stack and AI driven applications end to end using **Python**, **FastAPI**, Node.js, React, **PostgreSQL**, and **AWS**, delivering production quality features under weekly release cycles.
- Integrated AI ML features such as LLM powered components, inference workflows, and data processing services into application architectures to ensure stable and predictable execution.
- Implemented model driven features using **PyTorch** and **scikit learn** in Python for training, evaluation, and error analysis as part of applied ML project work.
- Led peer code reviews, provided PR level feedback, and enforced coding practices that improved code quality and reduced debugging time.
- Designed internal utilities, debugging scripts, and deployment helpers that accelerated development velocity for multi week team projects.
- Completed recurring engineering challenges with strict deadlines that replicated real production sprints, consistently ranking as a top performer.

Software Engineer Intern, University of Massachusetts Boston 01/2024 – 06/2024 | Boston, USA

- Designed a decoupled architecture for a patient guidance platform, separating the React frontend from the Python inference backend to enable independent scaling.
- Implemented a hybrid search engine combining vector-based retrieval with metadata rules to structure nursing content, reducing LLM hallucinations in patient facing responses.
- Engineered authentication and state management services to ensure consistent behavior across distributed message sessions.

Projects

PeerGenius - AI-powered student collaboration platform

- Engineered context aware AI response system with 95% accuracy in academic query detection, implementing logic to differentiate solo vs group study modes and delivering fast AI responses through Firebase Authentication and JWT security.
- Developed a full stack architecture with React 19 and Vite frontend, Express and MongoDB backend, Firebase Auth integration, and Radix UI component system, deployed with CI CD pipeline and production security standards.

LifeLens - Microservices-based AI daily decision engine

- Architected a full-stack microservices platform with Node.js and TypeScript backend, React Native mobile app, and FastAPI ML service, integrating 5+ AI APIs (OpenRouter, Google Vision, DistilBERT) with fallback mechanisms, achieving 95%+ uptime.
- Engineered production **PostgreSQL** (Row Level Security, optimized indexing) and Redis caching, boosting performance by 30% to 50% via token optimization and cache invalidation with sub 200ms query times.
- Built a contextual AI nudge engine processing 7+ real time factors (mood, weather, history) with a rule based system, reducing user decision fatigue.

Education

University of Massachusetts Boston | Boston, MA, USA Aug 2022 - May 2024

Master of Science in Computer Science

Sant Gadge Baba Amravati University | Maharashtra, India June 2019 - May 2022

Bachelor of Engineering in Computer Science and Engineering

Awards and Recognitions

- Boston Bridge Hackathon 2024 Champion**, University of Massachusetts Boston: Built My Beacon Bot, an AI assistant to improve commuter integration and deliver a personalized student support experience.